Richard (Jingbo) Gao

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Education

University of Chicago — Chicago, IL
B.S. Computer Science and B.S. Mathematics (Intended) — *Expected June 2029*

University of Virginia — Charlottesville, VA
Attended, transferred to UChicago — *Aug 2025 - Jul 2026, GPA: 3.955*

Experience

Undergraduate Research Assistant & Summer Researcher — Dec. 2025 – Present
VIVA Lab (Prof. Scott Acton) – University of Virginia — *Charlottesville, VA*

- Built a **training-free video face-anonymization pipeline** that de-identifies faces in classroom footage while preserving pose, gaze, and expression for activity analysis
- Integrated a diffusion de-identification model NullFace with YOLO-face detection, running the model as a persistent GPU worker across isolated conda environments
- Designed an identity embedding freezing mechanism with a similarity cache for consistent identities across frames.
- Deployed and benchmarked the system on UVA's Rivanna HPC cluster via SLURM GPU jobs across interview and classroom datasets
- Conducted literature review on diffusion and image/video anonymization, synthesized findings and presented to the lab

Data Analyst Intern — Dec. 2025 – Jan. 2026
Msunhealth — *Jinan, China and Remote*

- Analyzed **10k+ outpatient visit records** covering **9k patients** across **3 hospitals** using Python and MySQL
- Designed SQL queries and Python data-processing workflows to identify high-risk digestive disease and gastric cancer patients based on visit history and diagnosis codes
- Computed hospital-level screening metrics and visit-pattern statistics to evaluate screening effectiveness
- Produced analytical reports supporting data-driven clinical quality improvement initiatives

Student Developer — Sep. 2024 – Feb. 2025
Keystone Academy — *Beijing, China*

- Developed a Java application with JavaFX to generate, manage, and store formatted academic citations
- Implemented file I/O, sorting logic, and data persistence for real-world classroom use
- Version-controlled the project with Git and GitHub; software was adopted by classmates

Projects

Classroom Video Anonymization Pipeline | *Python, Bash*

- Built a training-free pipeline integrating YOLO face detection and diffusion de-identification methods.
- Improved identity consistency and visual quality using identity embedding cosine-similarity caching.

High-Risk Patient Identification & Screening Analysis | *Python, SQL, pandas*

- Engineered visit-frequency and diagnosis-based labeling rules using SQL queries and pandas transformations
- Generated hospital-level prevalence and screening metrics to support comparative analysis

Technical Skills

Languages: Python, Java, C++, SQL, R, Bash
Data & ML: PyTorch, NumPy, Pandas, Matplotlib, OpenCV
Tools: Git, Linux, Conda, Slurm, Markdown, LaTeX